

Backend Development Roadmap

Tools: Java(Springboot) syllabus

Core Java concepts

- classes/objects
- Inheritance,abstraction,polymorphism,encapsulation
- Exception handling
- Packages and Access Modifiers
- Java collection frameworks
- Generics
- String handling
- Wrapper classes

Advanced Java Concepts

- Java Memory Management
- JVM, JRE, JDK
- Garbage Collection
- Multithreading
 - Threads
 - Runnable
 - Executor Framework
 - Synchronization
- Java 8+ Features
 - Lambda Expressions
 - Stream API
 - Functional Interfaces
 - Optional
 - Date and Time API

Data Structure Algorithms knowledge

- Array,linkedlist,Stack,Queue,Hashtables,Trees,Sorting algorithms,searching algorithms, time and space complexity.
- Idea of when to use which stack according to optimality and considering time and space complexity factors
- Writting clean and optimized code, debugging techniques, understanding existing codebase and reading documentation.

Introduction & Environment Setup

- Spring Framework vs. Spring Boot
- Development Environment
- Spring Initializr
- Project Structure

Spring Boot Core Concepts

- Starters & Auto-configuration
- Spring Beans & Components
- Configuration & Profiles

Building RESTful Web APIs

- MVC Architecture
- Request Handling
- Responses & Status Codes
- Validation

Data Access & Persistence

- Spring Data JPA
- Database Configuration
- Repositories
- Entities & Relationships.

Exception Handling & Logging

- Global Exception Handling
- Logging Framework

Testing in Spring Boot

- Unit Testing
- Integration Testing

Security (Authentication & Authorization)

- Spring Security Basics
- Custom Configuration
- Modern Authentication

Production-Ready Features & Deployment

- Spring Boot Actuator
- Containerization
- Deployment.

Microservice Architecture

- Service discovery
- API Gateway
- Configuration server
- Distributed tracing
- Circuit breakers
- Fault tolerance

Spring Ecosystem

- Spring Cloud
- Spring Cloud Gateway
- Eureka
- OpenFeign
- Resilience4j

Build & Tooling

- Maven / Gradle
- Git & GitHub
- IDE Debugging Tools

Databases

- SQL Fundamentals (joins, indexing, normalization)
- PostgreSQL / MySQL
- Transaction Management
- Connection Pooling (HikariCP)
- NoSQL (MongoDB / Redis)

API Design

- API Versioning & Pagination
- Swagger / OpenAPI Documentation
- Idempotency & Rate Limiting

- GraphQL Basics

Security

- JWT & OAuth2
- CORS & CSRF
- Role-Based Access Control (RBAC)

Caching & Performance

- Redis / Ehcache
- Spring Cache Abstraction
- Async Processing (@Async)

Messaging & Event-Driven Systems

- Kafka
- RabbitMQ

Testing

- JUnit 5 & Mockito
- TestContainers
- Contract Testing

DevOps & Deployment

- Docker
- CI/CD (Jenkins / GitHub Actions)
- Cloud Basics (AWS/Azure/GCP)
- Monitoring (Prometheus, Grafana, ELK)

System Design & Patterns

- Load Balancing & Scaling
- Design Patterns (Singleton, Factory, Builder, Strategy)
- Agile/Scrum Basics

